

Sep. 21<sup>st</sup> 2026 Week 3 Paper-to-Peer

# **Beyond Geometry: Comparing the Temporal Structure of Computation in Neural Circuits with Dynamical Similarity Analysis**

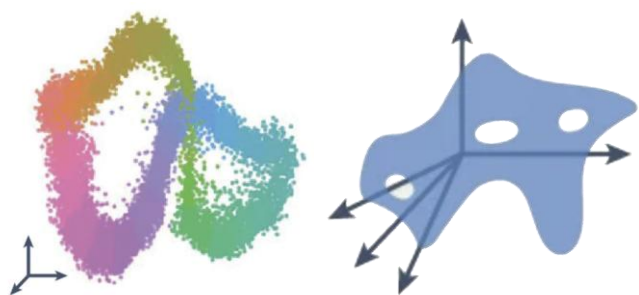
(Ostrow et al., 2023)

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# Motivation

How can we tell whether two dynamical systems utilize the *same internal processes* for a particular computation?

## Conventional similarity measures



Topology / Geometry / Dynamics

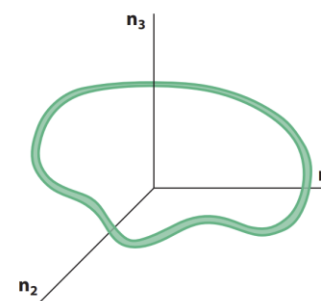
They mostly compared( $R^2$ ) the *geometry* of states in the latent space\* of a neural network.

- **Fixed point manifolds**

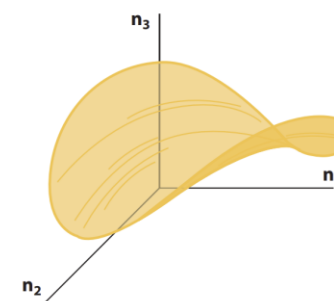
Fixed points are points in state space where repeated iteration of  $f$  does not change the state. Perturbations away from the fixed points collapse back to the fixed points.

$$\frac{dx}{dt} = f(x(t)) \Rightarrow f(x^*(t)) = 0$$

**b** 1D manifold ring



**c** 2D manifold

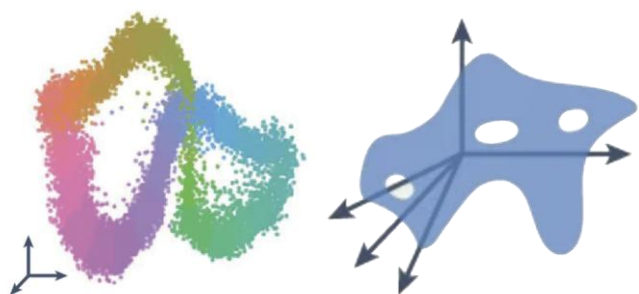


\* **Latent space:** low-dimensional mathematical space where high-dimensional data is compressed to capture its core hidden features

# Motivation

How can we tell whether two dynamical systems utilize the *same internal processes* for a particular computation?

## Conventional similarity measures



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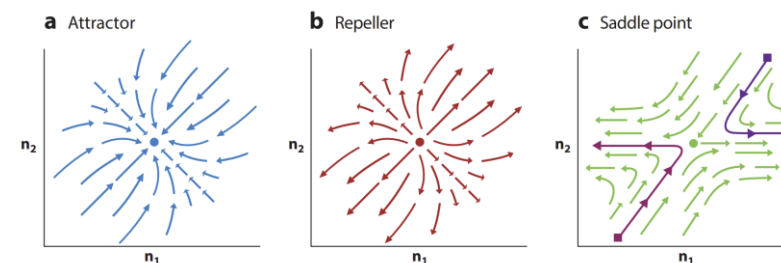
They mostly compared( $R^2$ ) the *geometry* of states in the latent space\* of a neural network.

But...

- Geometries can differ due to sampling differences or other fundamental causes even though the dynamics are *topologically equivalent*  
→ False negative



- System may exhibit *distinct dynamics* over *similar geometries* in the state space  
→ False positive



Computations are implemented at the level of *dynamics*!  
→ **Dynamical Similarity Analysis (DSA)**

# Some Math

## How can we define “being the same”?

### Step 1

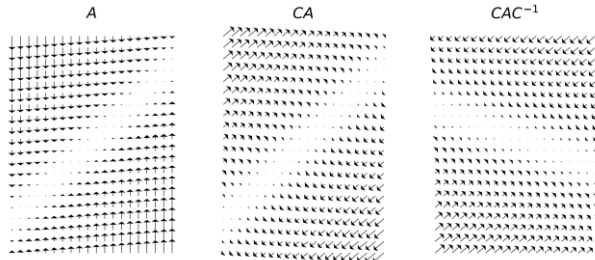
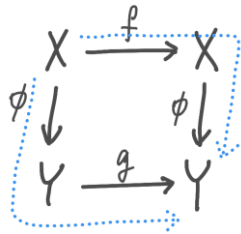
#### Conjugacy

In a group  $G$ ,  $a, b \in G$  are conjugate if  $\exists g \in G$  s.t.  $a = gb g^{-1}$

#### Topological Conjugacy

Given two dynamical systems  $f: X \rightarrow X$  and  $g: Y \rightarrow Y$  with invertible mapping  $\phi: X \rightarrow Y$ , conjugacy is defined as

$$g \circ \phi = \phi \circ f$$



→ Coordinate conversion from a system to another.  
If conjugate, two systems have the **same qualitative behavior**.

### Step 2

#### Koopman Operator

For a discrete-time dynamical system  $x_{t+1} = F(x_t)$ , the Koopman operator  $K$  is defined by

$$K \circ g = g \circ F$$

- $g$  is an “observable”:  
a function that maps the state to a scalar  $g: X \rightarrow \mathbb{R}$

$$g_1(x) = x, g_2(x) = \sin(x) \\ \Rightarrow g(x) = [g_1(x) \ g_2(x) \ \dots \ g_m(x)]^T$$

- $K$  is a linear operator:

$$K(\alpha g + \beta h) = (\alpha g + \beta h) \circ F = \alpha(g \circ F) + \beta(h \circ F)$$

It simplifies the problem into  $g_{t+1} = K g_t$  ( $g_t = g(x_t)$ ).

→ Enables modeling of nonlinear dynamics into **linear systems**, which are more tractable & interpretable

# Some Math

## How can we define “being the same”?

### Step 3

#### Dynamic Mode Decomposition (DMD)

DMD identifies a linear transition operator for the dynamics:

$$X(t + \Delta t) = AX(t)$$

$$A_{DMD} = X_{t+\Delta t} X_t^\dagger = X_{t+\Delta t} V \Sigma^\dagger U^T$$

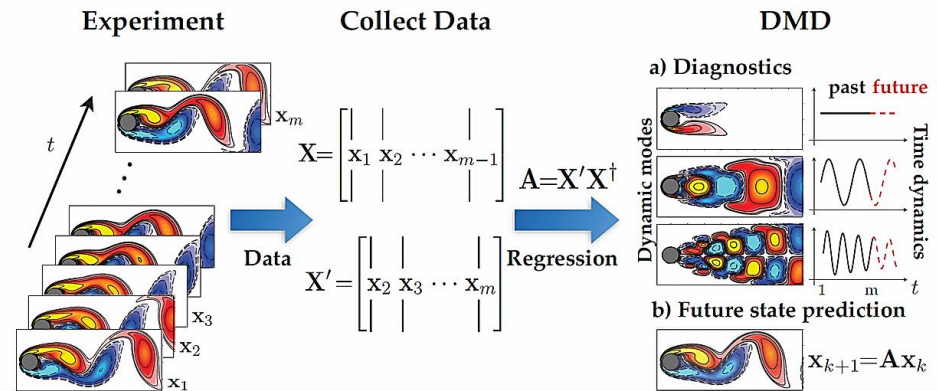
$$\Rightarrow x_n = \sum_{i=1}^M c_i \lambda_i^k v_i \quad (A v_i = \mu_i v_i \quad \forall i = 1, \dots, M)$$

- Koopman operator is theoretically infinite-dimension:

$$g_{t+1} = K g_t \stackrel{?}{\Rightarrow} K = g_{t+1} g_t^{-1}$$

$$G = \{g: \mathbb{R} \rightarrow \mathbb{R}\} \quad (\text{e.g., } 1, x, x^2, x^4, \dots)$$

→ Can use to set a **finite-dimensional** embedding that approximate the Koopman operator



#### Hankel Matrix

$$X = [-x(t) \quad -x(t - \tau) \quad \dots \quad -x(t - p\tau) \quad -]^T$$

$$\text{e.g., } X = \begin{bmatrix} x(t) & x(t + \tau) & x(t + 2\tau) & \dots \\ x(t - \tau) & x(t) & x(t + \tau) & \dots \\ x(t - 2\tau) & x(t - \tau) & x(t) & \dots \end{bmatrix}$$

- $X$  is closed under the operation of  $A$ :  $X \in G \Rightarrow AX \in G$
- Improves estimation capabilities over partially-observed systems

# Some Math

How can we define “being the same”?

- **Comparing Koopman operator reflects dynamical similarity.**

**Theorem 3** (Spectral equivalence of topologically conjugate systems Budišić et al. [2012]). *Let  $S : M \rightarrow M$  and  $T : N \rightarrow N$  be maps that define two topologically conjugate dynamical systems over different manifolds  $M$  and  $N$ . Let there exist a homeomorphism  $h : N \rightarrow M$  such that  $S \circ h = h \circ T$ . If  $\phi$  is an eigenfunction of the Koopman operator of  $S$ ,  $U_s$  (i.e.  $U_s \phi = \lambda \phi$ ), then  $\phi \circ h$  is an eigenfunction of the Koopman operator of  $T$ ,  $U_t$  also with eigenvalue  $\lambda$ .*

*Proof.* Define  $x \in M$  and let  $y \in N$  such that  $x = h(y)$ . Then

$$\begin{aligned} U_S \phi(x) &= \phi(S(x)) = \lambda \phi(x) \\ \phi \circ (S \circ h)(y) &= \lambda(\phi \circ h)(y) \\ \phi \circ (h \circ T)(y) &= \lambda(\phi \circ h)(y) \\ U_T(\phi \circ h)(y) &= \lambda(\phi \circ h)(y) \end{aligned}$$

□

# Some Math

## How can we define similarity?

### Procrustes Analysis over Vector Fields

A shape metric which compares the geometric similarity of two datasets after identifying an optimal orthogonal transformation\* to align them.

$$d(\mathbf{A}_x, \mathbf{A}_y) = \min_{\mathbf{C} \in O(n)} \|\mathbf{A}_x - \mathbf{C}\mathbf{A}_y\mathbf{C}^{-1}\|_F$$
$$\Rightarrow \min_{\mathbf{C} \in O(n)} \arccos \frac{\mathbf{A}_x \cdot (\mathbf{C}\mathbf{A}_y\mathbf{C}^{-1})}{|\mathbf{A}_x|_F |\mathbf{A}_y|_F}$$

- Zero:  $d(A, B) = 0 \Leftrightarrow A = B$
- Symmetry:  $d(A, B) = d(B, A)$
- Triangle Inequality:  $d(A, C) \leq d(A, B) + d(B, C)$

- $d(A, B)$  is a metric.

$g(A, B) = \|A - B\|_F$  is a metric (trivially true) & the similarity transform is an isometry

$$g(T(A), T(B)) = g(A, B)$$

$$g(T(A), T(B)) = \|CAC^{-1} - CBC^{-1}\|_F = \|C(A - B)C^{-1}\|_F$$

$$g(T(A), T(B)) = \text{Tr}([C(A - B)C^{-1}]^T [C(A - B)C^{-1}])$$

$$= \text{Tr}(C(A - B)^T C^{-1} C(A - B) C^{-1})$$

$$= \text{Tr}(C(A - B)^T (A - B) C^{-1}) = \text{Tr}(C^{-1} C(A - B)^T (A - B))$$

$$= \text{Tr}((A - B)^T (A - B)) = \|A - B\|_F = g(A, B)$$

→ If the transformations do not distort the original metric, then we can define a consistent distance between the transformed objects

\*  $O(n)$ : The group of orthogonal matrices of dimension  $n$ . Refers to rotation if  $\det(A) = 1$ , and reflection if  $\det(A) = -1$ .

# Some Math

How can we define similarity?

- **Euclidean distance to angular distance**

$$\begin{aligned}\|A - CBC^{-1}\|^2 &= \text{Tr}((A - CBC^{-1})^T(A - CBC^{-1})) \\ &= \text{Tr}(A^T A) - 2\text{Tr}(A^T CBC^{-1}) + \text{Tr}(B^T B) \\ &= \text{Tr}(A^T A) - \boxed{2 \langle A, CBC^{-1} \rangle} + \text{Tr}(B^T B) \\ &\quad \rightarrow \text{maximize} \\ \text{arccos} \frac{\langle A, CBC^{-1} \rangle}{\|A\| \|B\|}\end{aligned}$$

→ Since arccos function monotonically decreases, maximizing the matrix inner product minimizes the angular distance.

# Some Math

## How can we measure similarity?

### Step 1

$C \in SO(n)$  covers all the possible cases

We cannot continuously parametrize all of  $O(n)$ , as it is made up of two disconnected components  $SO(n)$  where  $\det(C) = 1$  and  $O(n) \setminus SO(n)^*$ .

**Theorem 1.** All elements of  $O(n)/SO(n)$  are bijective maps between itself and  $SO(n)$ .

*Proof.* Let  $D, P \in O(n)/SO(n)$ . Then  $PD \in SO(n)$ . For this to be true,  $PD$  must have two properties:  $\det(PD) = 1$ , and  $(PD)^T(PD) = I$ . The first is easily shown:  $\det(PD) = \det(P)\det(D) = -1 * -1 = 1$ . The second is likewise shown easily:  $(PD)^T(PD) = D^T P^T P D = D^T D = I$  by orthogonality. Both  $P$  and  $D$  have inverses  $P^T$  and  $D^T$ .

Conversely, let  $D \in SO(n)$ . Then we can show by the same logic that  $PD \in O(n)/SO(n)$ .  $\square$

$$\Rightarrow \min_{Q \in O(n)} \|B - QAQ^T\|_F = \min \left\{ \begin{array}{l} \min_{C \in SO(n)} \|B - CAC^T\|_F, \\ \min_{C \in SO(n)} \|B - CPA P^T C^T\|_F \end{array} \right\}$$

\*  $SO(n)$  is a subgroup of  $O(n)$  while  $O(n) \setminus SO(n)$  is not, as the latter does not have the identity element.

# Some Math

## How can we measure similarity?

### Step 2

#### Cayley Transform

For an arbitrary skew-matrix\*  $\mathbf{S}$ , the matrix  $\mathbf{C}$  calculated as below is orthogonal:

$$\mathbf{C} = (\mathbf{I} - \mathbf{S})(\mathbf{I} + \mathbf{S})^{-1}$$

- Cayley Transform restricts the operation to  $SO(n)$  :

$$\begin{aligned}\mathbf{C} &= (\mathbf{I} - \mathbf{S})(\mathbf{I} + \mathbf{S})^{-1} \Rightarrow \mathbf{C}^T = [(\mathbf{I} + \mathbf{S})^{-1}]^T (\mathbf{I} - \mathbf{S})^T = [(\mathbf{I} + \mathbf{S})^T]^{-1} (\mathbf{I} - \mathbf{S})^T \\ &\Rightarrow \mathbf{C}^T = (\mathbf{I} - \mathbf{S})^{-1} (\mathbf{I} + \mathbf{S}) \\ &\Rightarrow \mathbf{C}^T \mathbf{C} = (\mathbf{I} - \mathbf{S})^{-1} (\mathbf{I} + \mathbf{S})(\mathbf{I} + \mathbf{S})^{-1} (\mathbf{I} - \mathbf{S}) = \mathbf{I}\end{aligned}$$

$$\det(\mathbf{C}) = \det[(\mathbf{I} - \mathbf{S})(\mathbf{I} + \mathbf{S})^{-1}] = \frac{\det(\mathbf{I} - \mathbf{S})}{\det(\mathbf{I} + \mathbf{S})} = \frac{\det((\mathbf{I} - \mathbf{S})^T)}{\det(\mathbf{I} + \mathbf{S})} = 1$$

→ Re-parametrization allows computing the metric with gradient descent over  $\mathbf{S}$  via Adam optimizer (e.g., `C = torch.linalg.solve(I+S)(I-S)`).

\* **Skew-matrix:** Matrices that satisfy  $X^T = -X$ .

# Results

## DSA captures dynamical similarity and differences

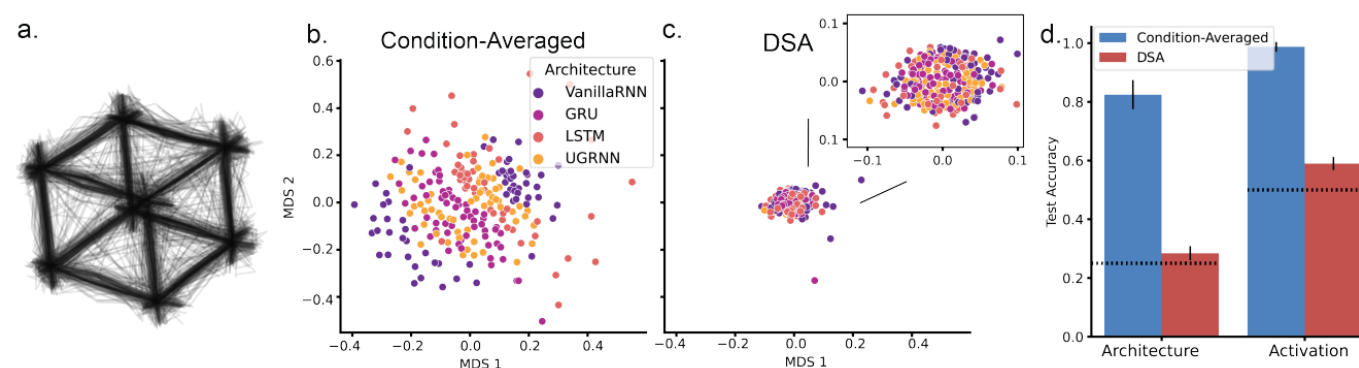


Figure 2: Same dynamics, different shape, only identified as similar with DSA.

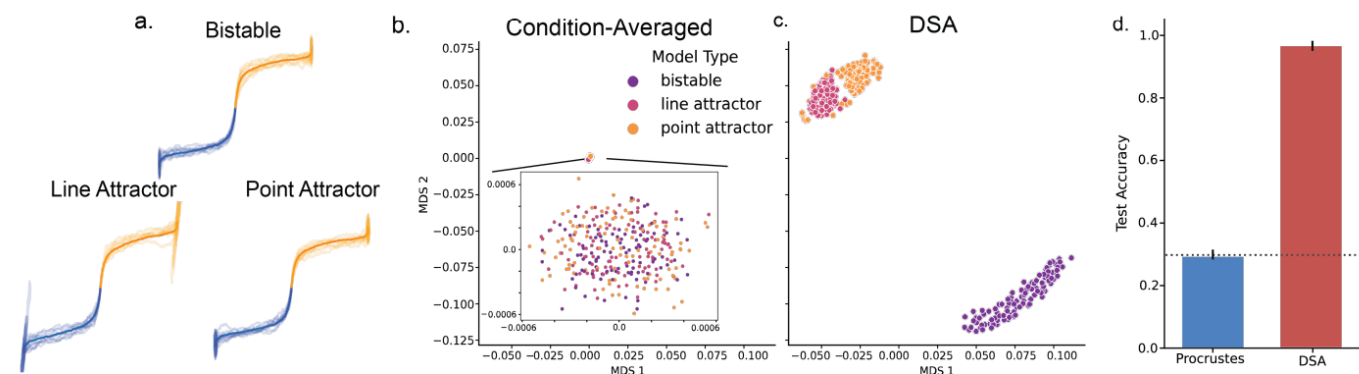


Figure 3: Different dynamics, same shape, only distinguished by DSA.

# Results

**DSA responds to topology changes, but does not to geometric deformations**

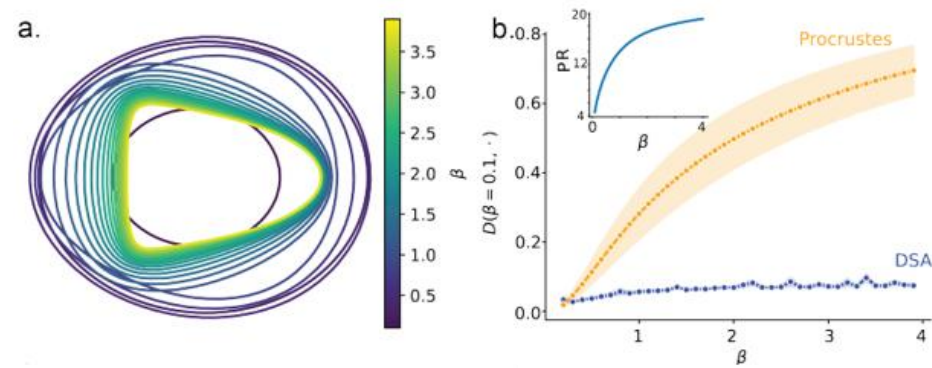


Figure 4a: DSA is invariant under geometric deformation of a ring attractor.

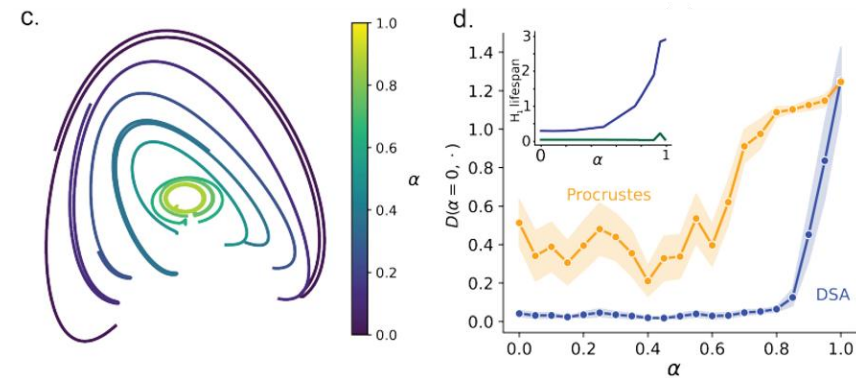


Figure 4b: DSA is sensitive to the transformation of a ring attractor into a line.

# Results

DSA can disentangle learning rules in an unsupervised fashion

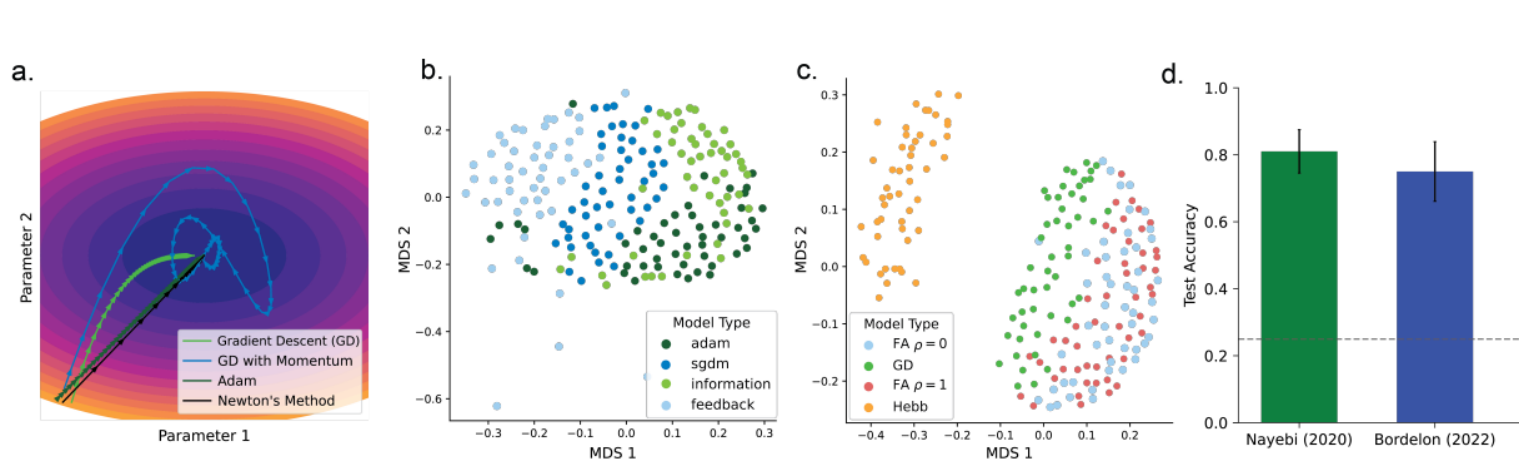


Figure 5: DSA unsupervisedly disentangles learning rules.

**Thank you 😊**

# Appendix

## References

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